

Ahmed Arif

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EDUCATION

CORNELL UNIVERSITY

Ithaca, NY

B.S. in Mechanical and Aerospace Engineering

Expected May 2028

RELEVANT COURSEWORK: Calculus I–III, Linear Algebra, Differential Equations, Dynamics, Statics & Mechanics of Solids, Thermodynamics, Mechanical Design, Physics I & E&M, Fluid Mechanics, Python

SKILLS

- Simulation & Analysis: ANSYS Fluent: meshing, k- ω SST turbulent, transient, VOF multiphase, compressible, CHT, DPM; ANSYS Mechanical: static structural, steady-state & transient thermal; OpenRocket, RPA
- Technical skills: Python, Autodesk Fusion 360, Inventor, and Vault, Solidworks, Bambu Lab Studio, Arduino

ENGINEERING EXPERIENCE

LIQUID PROPULSION AT CORNELL

Ursa I Subteam Analyst

November 2025 - June 2026

- Conducted multi-fidelity CFD in ANSYS Fluent progressing from 2D laminar/turbulent (k- ω SST) to 3D incompressible VOF multiphase transient simulations modeling ethanol/water cold flow with an air initialization phase to characterize impinging injector performance of the 500 N N₂O/E99 Ursa I bipropellant engine
- Post-processed velocity, pressure, and volume fraction contours and volume renderings to map recirculation zones at the impingement point and extract discharge coefficient; identified discrepancy (in 2D) between CFD-predicted and analytical Cd, indicating the need for 3D injector geometry to improve model fidelity
- Validated manifold pressure stability across operating conditions; designed load cell spacer to convert hot-fire test data into structured inputs for Ursa II engine development

CORNELL MARS ROVER PROJECT TEAM

Ithaca, NY

Arm Subteam Engineer

September 2025 - Present

- Revised the end effector and swappable tool kit from documented failure modes of the previous competition robot (broken key clicker, compliant silicone, impact damage), targeting grip strength and a more compact design
- Bench-tested a 3D-printed skeleton intended to stiffen the silicone grips, using an Arduino servo controller and Python script on five objects up to 5.9 kg; objects rotated in the claw with and without it, so now evaluating silicone grade
- Iterated the design through two silicone mold revisions and added hard stops to keep the claws from rotating when closed; designed the mounting plate for the arm's new CubeMars motor and previously, a 2-DOF camera gimbal

ENGINEERING PROJECTS

Liquid Rocket Engine Design

May 2026 – Present

- Designed a 5 kN LOX/RP-1 liquid rocket engine from first principles using RPA and NASA CEA for thermochemical analysis, defining full engine sizing and an 80% bell nozzle at 400 psi chamber pressure
- Designed an impinging injector (unlike-triplet elements, 45° impingement angle) and validated impingement behavior via multiphase CFD in ANSYS Fluent; injector-level transient thermal FEA revealed insufficient dome cooling and excessive faceplate temperatures despite Inconel construction, prompting a custom passive LOX bleed cooling system for the dome
- Designed and sized regenerative cooling channels for the chamber and nozzle based on thermal margin and manufacturability constraints, following initial transient thermal FEA showing uncooled chamber failure in under one second
- Currently running a non-premixed combustion CHT with DPM simulation incorporating regenerative cooling channel geometry and a hot-gas inlet approximating future ASI integration; confirmed correct DPM particle behavior and am testing realizable k- ϵ turbulence modeling to improve convergence

ROVER PATHFINDING SIMULATION

June 2025

- Developed a Python [pathfinding simulation](#) from scratch which navigates a rover on a multi-terrain grid to reach its target in the shortest amount of time (with .001 tile accuracy) using a custom function for Dijkstra's Algorithm
- Programmed a logging-system that displays various attributes of the rover for every step in real time, such as position and progress, for the entirety of the rover's journey

VOLUNTEER/LEADERSHIP EXPERIENCE

MIDDLETOWN ISLAMIC CENTER WEEKEND SCHOOL

New Hampton, NY

Volunteer Teacher

September 2019 - June 2024

- Created and adapted lessons for 15+ students weekly to facilitate an effective educational environment
- Fostered student interest and curiosity through engaging classroom activities